Correction TD thermodynamique

Exercice 1:

1) transformation isotherme : $W = \int_{V_1}^{V_2} -P\,dV$ or on a $PV = nRT$

$P = \frac{nRT}{V}$

$\Rightarrow W = -\int_{V_1}^{V_2} \frac{nRT}{V}\,dV = -nRT\int_{V_1}^{V_2} \frac{1}{V}\,dV$

$= -nRT\,[\ln V]_{V_1}^{V_2} = -nRT\,[\ln V_2 - \ln V_1]$

$\Rightarrow \boxed{W = -nRT\,\ln\left(\frac{V_2}{V_1}\right)}$

Pour $V_2 > V_1 \Rightarrow \frac{V_2}{V_1} > 1 \Rightarrow \ln\left(\frac{V_2}{V_1}\right) > 0 \Rightarrow W < 0$.

$V_2 < V_1 \Rightarrow \frac{V_2}{V_1} < 1 \Rightarrow \ln\left(\frac{V_2}{V_1}\right) < 0 \Rightarrow W > 0$

2) $PV = nRT$ or $n = \frac{m}{M} \Rightarrow \boxed{PV = \frac{mRT}{M}}$

3) $C_V = \frac{nR}{\gamma - 1}$; $C_P = \frac{nR\gamma}{\gamma - 1}$

4) La variation de l'energie interne d'un gaz parfait ne dépend que de la Température : $U = f(T) \Rightarrow \Delta U = C_V \cdot \Delta T$.

5) $\Delta U = U_f - U_i$: c'est une fonction d'état

Pour une transformation cyclique $U_f = U_i \Rightarrow \Delta U_{cycle} = 0$.

6) Pour la transformation isobare : $\Delta Q = \Delta H$

7) Pour une transformation isochore : $\Delta Q = \Delta U$.

8)
- $PV^{\gamma} = cte$
- $P^{1-\gamma}\,T^{\gamma} = cte$
- $TV^{\gamma-1} = cte$

Exercice 2 :

1) on a $P_1V_1 = nRT_1$; avec : $1\,bar = 10^5\,Pa$

$$\Rightarrow V_1 = \frac{nRT_1}{P_1} = \frac{1 \times 8{,}32 \times 330}{10^5} = 0{,}027\,m^3$$

$$\boxed{V_1 = 0{,}027\,m^3}$$

2) A_1A_2 : adiabatique : $PV^\gamma = cte$

$$P_2V_2^\gamma = P_1V_1^\gamma \Rightarrow P_2 = \left(\frac{V_1}{V_2}\right)^\gamma \cdot P_1 = x^\gamma \cdot P_1 = (14)^{1,4} \times 10^5$$

$$\boxed{P_2 = 40{,}23\,10^5\,Pa}$$

3) $$P_2V_2 = nRT_2 \Rightarrow T_2 = \frac{P_2V_2}{nR} = \frac{P_2 \cdot \left(\frac{V_1}{x}\right)}{nR} = \frac{40{,}23\,10^5 \cdot \frac{0{,}027}{14}}{1 \times 8{,}32}$$

$$\boxed{T_2 = 932{,}52\,K}$$

En fin de Combustion, la température $T_3 = 2260\,K$.

1) $P_3V_3 = nRT_3 \Rightarrow V_3 = \frac{nRT_3}{P_3}$ ($P_3 = P_2$: A_2A_3 est isobare)

$$V_3 = \frac{1 \times 8{,}32 \times 2260}{40{,}23 \cdot 10^5} = 4{,}67\,10^{-3}\,m^3.$$

2) A_2A_3 : isobare $\Rightarrow Q_{23} = nC_p\Delta T = nC_p(T_3 - T_2)$
$= 1 \times 29\,(2260 - 932{,}52)$

$$\boxed{Q_{23} = 38{,}49\,10^3\,J}$$

3) A_3A_4 adiabatique : $P_4V_4^\gamma = P_3V_3^\gamma$

$\Rightarrow P_4 = \left(\frac{V_3}{V_4}\right)^\gamma P_3$ or $V_4 = V_1$ A_4A_1 : isochore

$$P_4 = \left(\frac{4{,}67\,10^{-3}}{0{,}027}\right)^{1,4} \times 40{,}23\,10^5 = 3{,}48\,10^5\,Pa.$$

4) $$P_4V_4 = nRT_4 \Rightarrow T_4 = \frac{P_4V_4}{nR} = \frac{3{,}48\,10^5 \times 0{,}027}{1 \times 8{,}32}$$

$$\boxed{T_4 = 1129{,}32\,K}$$

5) A_4A_1 : isochore : $Q_{41} = nC_v\Delta T = nC_v(T_1 - T_4)$

$$Q_{41} = n\frac{C_p}{\gamma}(T_1 - T_4)$$

$$= \frac{29}{1,4}(330 - 1129,32) = -16557\,J$$

6) $\Delta U_{cy} = W_{cy} + Q_{cy} = 0\,J$

$$\Rightarrow W_{cy} = -Q_{cy} = -(Q_{12} + Q_{23} + Q_{34} + Q_{41})$$

$$= -(0 + 38,49\,10^3 + 0 - 16,557\,10^3)$$

$$W_{totale} = W_{cy} = -21,93\,10^3\,J$$

7) $\eta = \left|\frac{W}{Q_{23}}\right| = \left|\frac{-21,93\,10^3}{38,49\,10^3}\right| = 0,56 = 56\,\%$

Exercice 3 :

1) • $P_A V_A = nRT_A \Rightarrow V_A = \frac{nRT_A}{P_A} = \frac{1 \times 8,32 \times 298}{10^5}$

$$\boxed{V_A = 0,025\,m^3}$$

• $A \longrightarrow B$: adiabatique : $P_A V_A^\gamma = P_B V_B^\gamma \Rightarrow V_B^\gamma = \frac{P_A V_A^\gamma}{P_B}$

$$V_B = \left(\frac{P_A V_A^\gamma}{P_B}\right)^{\frac{1}{\gamma}} = \left(\frac{10^5 (0,025)^{1,4}}{2,2\,10^5}\right)^{\frac{1}{1,4}} = 0,014\,m^3$$

$$\boxed{V_B = 0,014\,m^3}$$

• $P_B V_B = nRT_B \Rightarrow T_B = \frac{P_B V_B}{nR} = \frac{2,2\,10^5 \times 0,014}{1 \times 8,32} = 370,19\,K$

$$\boxed{T_B = 370,19\,K}$$

• $P_C = P_B = 2,2\,10^5\,Pa$ ($B \longrightarrow C$: transformation isobare)

• $P_C V_C = nRT_C \Rightarrow V_C = \frac{nRT_C}{P_C} = \frac{1 \times 8,32 \times 340}{2,2\,10^5} =$

$$\boxed{V_C = 0,013\,m^3}$$

• $P_D = P_A = 10^5\,Pa$ ($D \rightarrow A$: transformation isobare)

• $C \rightarrow D$: adiabatique : $P_C V_C^{\gamma} = P_D V_D^{\gamma}$

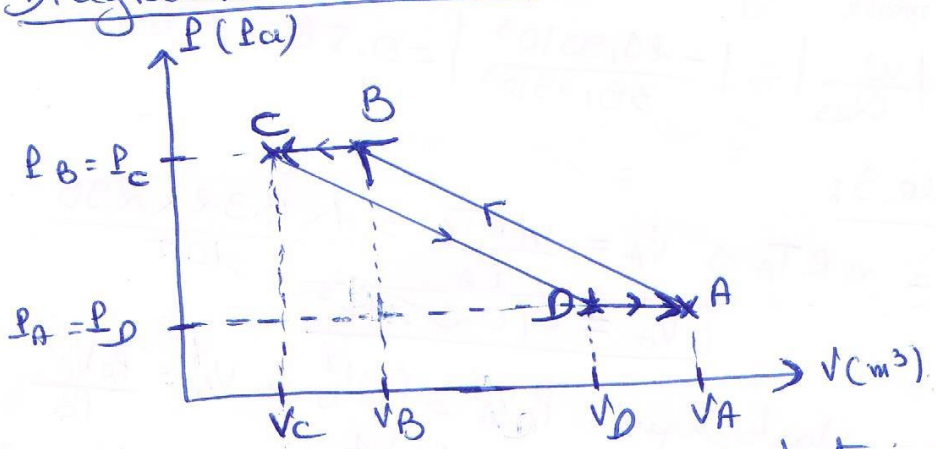
$$V_D^{\gamma} = \frac{P_C V_C^{\gamma}}{P_D} \Rightarrow V_D = \left(\frac{P_C \cdot V_C^{\gamma}}{P_D}\right)^{\frac{1}{\gamma}} = \left(\frac{2{,}2\,10^5 \cdot (0{,}013)^{1{,}4}}{10^5}\right)^{\frac{1}{1{,}4}}$$

$$\boxed{V_D = 0{,}023\ m^3}$$

• $P_D V_D = n R T_D \Rightarrow T_D = \frac{P_D V_D}{n R} = \frac{10^5 \cdot 0{,}023}{1 \times 8{,}32} = 276{,}4\,k$

$$\boxed{T_D \simeq 271\ k}$$

2) Diagramme de Clapeyron (P, V)



3). $Q_{AB} = 0\,J$ (transformation adiabatique).

• $Q_{BC} = n C_p \Delta T = n C_p (T_C - T_B) = 1 \times 29{,}1\,(340 - 370{,}19)$

$$\boxed{Q_{BC} = -878{,}53\ J}$$

• $Q_{CD} = 0\,J$

• $Q_{DA} = n C_p \Delta T = n C_p (T_A - T_D) = 1 \times 29{,}1\,(298 - 271)$

$$Q_{DA} = 785{,}7\ J$$

4) $\Delta U_{cy} = 0\,J \Rightarrow \Delta U_{cy} = Q_{cy} + W_{cy} = 0 \Rightarrow W_{cy} = -Q_{cy}$

$W_{cy} = -(Q_{AB} + Q_{BC} + Q_{CD} + Q_{DA}) = -(0 - 878{,}53 + 0 + 785{,}7)$

$$\boxed{W_{cy} = 92{,}83\,J}$$

5) $e = \frac{-Q_C}{W}$

$e = \frac{-Q_{BC}}{W_{ay}} = -\left(\frac{-878{,}53}{92{,}83}\right) = 9{,}46$

Exercice 4 :

1) • Etat 2 :

$1 \longrightarrow 2$: adiabatique : $P_2 V_2^{\gamma} = P_1 V_1^{\gamma} \Rightarrow P_2 = \frac{P_1 V_1^{\gamma}}{V_2^{\gamma}}$

$P_2 = P_1 \left(\frac{V_1}{V_2}\right)^{\gamma} \Rightarrow \boxed{P_2 = P_1 . a^{\gamma}}$

$\frac{V_1}{V_2} = a \Rightarrow \boxed{V_2 = \frac{V_1}{a}}$

$P_2 V_2 = nRT_2 \Rightarrow T_2 = \frac{P_2 V_2}{nR} = \frac{P_1 a^{\gamma} . \frac{V_1}{a}}{nR}$

or $P_1 V_1 = nRT_1 \Rightarrow nR = \frac{P_1 V_1}{T_1} \Rightarrow T_2 = \frac{\cancel{P_1} a^{\gamma} . \frac{\cancel{V_1}}{a}}{\frac{\cancel{P_1 V_1}}{T_1}}$

$T_2 = T_1 . a^{\gamma} . a^{-1} \Rightarrow \boxed{T_2 = a^{\gamma - 1} . T_1}$

• Etat 3 :

$2 \longrightarrow 3$: isobare : $\boxed{P_3 = P_2 = P_1 . a^{\gamma}}$

$b = \frac{V_4}{V_3} \Rightarrow V_3 = \frac{V_4}{b}$ or $4 \longrightarrow 1$: isochore : $V_4 = V_1$

$\Rightarrow \boxed{V_3 = \frac{V_1}{b}}$

$P_3 V_3 = nRT_3 \Rightarrow T_3 = \frac{P_3 V_3}{nR} = \frac{\cancel{P_1} . a^{\gamma} . \frac{\cancel{V_1}}{b}}{\frac{\cancel{P_1 V_1}}{T_1}} = T_1 a^{\gamma} . \frac{1}{b}$

$\boxed{T_3 = \frac{T_1 a^{\gamma}}{b}}$

Etat 4 :

$3 \longrightarrow 4$: adiabatique

$$P_3 V_3^{\gamma} = P_4 V_4^{\gamma} \Rightarrow P_4 = P_3 \cdot \left(\frac{V_3}{V_4}\right)^{\gamma} = P_1 a^{\gamma} \cdot \left(\frac{1}{b}\right)^{\gamma}$$

$$\Rightarrow \boxed{P_4 = P_1 \left(\frac{a}{b}\right)^{\gamma}}$$

$\boxed{V_4 = V_1}$: $4 \longrightarrow 1$: isochore

$$P_4 \cdot V_4 = nRT_4 \Rightarrow T_4 = \frac{P_4 V_4}{nR} = \frac{\cancel{P_1} \cdot \left(\frac{a}{b}\right)^{\gamma} \cdot \cancel{V_1}}{\frac{\cancel{P_1 \cdot V_1}}{T_1}}$$

$$\boxed{T_4 = T_1 \left(\frac{a}{b}\right)^{\gamma}}$$

2)

	P [Pa] 10^5	Volume [m³] 10^{-3}	Température [K]
Etat 1	1	$V_1 = \frac{nRT_1}{P_1} = \frac{1 \times 8{,}32 \cdot 300}{10^5} = 24$	300
Etat 2	25,1	2,4	753,6
Etat 3	25,1	4,8	1507,1
Etat 4	2,6	24	791,7

Exercice 5 : $R = C_p - C_v = 8{,}4\ J \cdot mol^{-1} \cdot K^{-1} \simeq 8{,}32$

1) $1 \longrightarrow 2$: isotherme

$$W_{12} = -nRT \ln\left(\frac{V_2}{V_1}\right)$$

on a $T_1 = T_2$ et $n = 0{,}04$ mole

$$W_{12} = -0{,}04 \times 8{,}32 \times 300 \ln\left(\frac{9{,}9 \cdot 10^{-5}}{10^{-5}}\right)$$

$\cdot\ P_2 V_2 = nRT_2$

$$\Rightarrow V_2 = \frac{nRT_2}{P_2} = \frac{0{,}04 \times 8{,}32 \times 300}{10 \cdot 10^5}$$

$$V_2 = 9{,}9\ 10^{-5}\ m^3$$

$$\boxed{W_{12} = 230{,}89\ J}$$

$$U_{isotherme} = 0\ J \Rightarrow U_{12} = 0\ J \Rightarrow W_{12} + Q_{12} = 0 \Rightarrow \boxed{Q_{12} = -230{,}89\ J}$$

$2 \longrightarrow 3$: isochore : $V_2 = V_3$

$W_{23} = 0\,J$; $Q_{23} = n c_v (T_3 - T_2)$

<u>calculons T_3</u>

$P_3 V_3 = n R T_3 \Rightarrow T_3 = \frac{P_3 V_3}{n R} = \frac{10^5 \times 9{,}9\,10^{-5}}{0{,}04 \times 0{,}32}$

$P_3 = P_1$ $\Rightarrow T_3 = 29{,}7\,K$.

$Q_{23} = 0{,}04 \times 21 \times (29{,}7 - 300)$

$Q_{23} = -227{,}05\,J$

$3 \longrightarrow 1$: isobare

$W_{31} = -P_3 \Delta V = -P_3 (V_1 - V_3) = -10^5 (10^{-3} - 9{,}9\,10^{-5})$

$= -90{,}1\,J$.

$\boxed{W_{31} = -90{,}1\,J}$

$Q_{31} = n c_p \Delta T = 0{,}04 \times 29{,}4 (300 - 29{,}7)$

$= n c_p (T_1 - T_3) = 317{,}8\,J$

$\boxed{Q_{31} = 317{,}8\,J}$

2) $U_{cycle} = U_{12} + U_{23} + U_{31} = (W_{12} + Q_{12}) + (W_{23} + Q_{23}) + (W_{31} + Q_{31})$

$= (230{,}89 - 230{,}89) + (0 - 227{,}05) + (-90{,}1 + 317{,}8)$

$= 0{,}65\,J \simeq 0\,J$.